Titanic Analysis Report.

Introduction

On April 15, 1912, the RMS Titanic sank claiming 1,502 lives. This report presents an exploratory data analysis of 891 passengers to identify the demographic factors that most significantly influenced survival probability. For this analysis, a dataset including the total number of passengers, their names, boarding class, gender, age along with other information will be used to understand the various survival rates of different demographics of people within the titanic such as the females as compared to males, youthful as compared to elderly, etc.

Data Overview

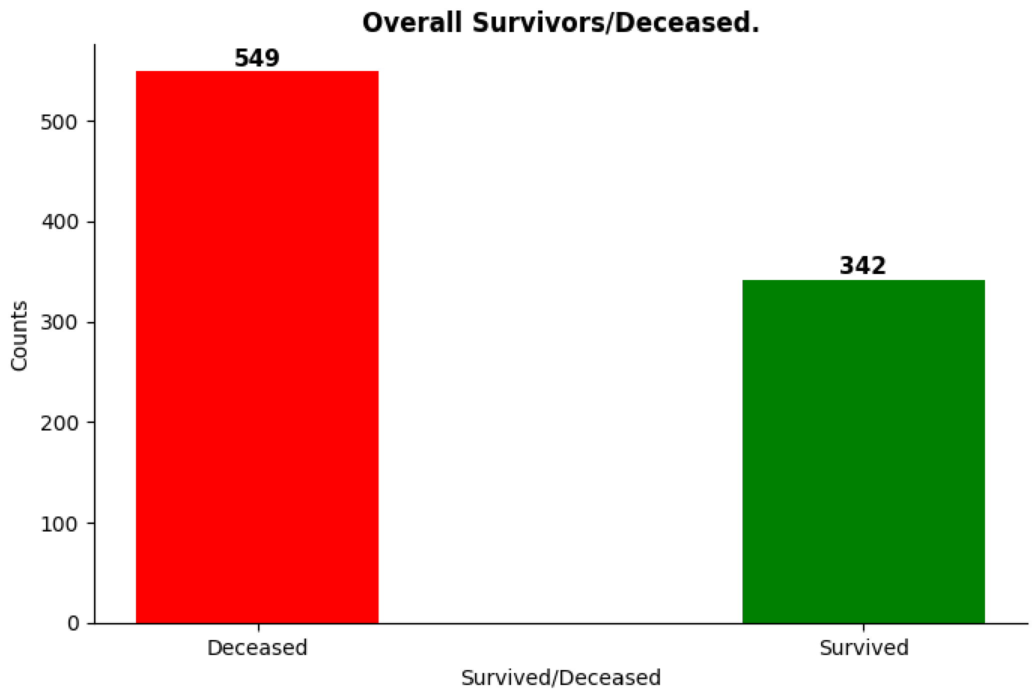
The titanic dataset includes 891 passengers and 12 columns. the key data quality issues include three columns with missing data entry points namely the; 'embarked', 'age', and 'cabin' column. The embarked column includes the list amount of data entry points with only 2 missing. The 'Age' column however possesses 177 missing data points accounting for 20% of its total data, while the 'cabin' column includes the highest number of missing entries having 687 null data entries which accounts for 77% of its total data. For the sake of clean analysis, the null data for each missing data entry would be filled for both the 'Age' and 'embarked' column, while the 'cabin' column will be dropped entirely. as for the Age column, the null values will be filled with the median age; while the 'embarked' column will be filled with the most occurring data entry.

Key findings

Overview

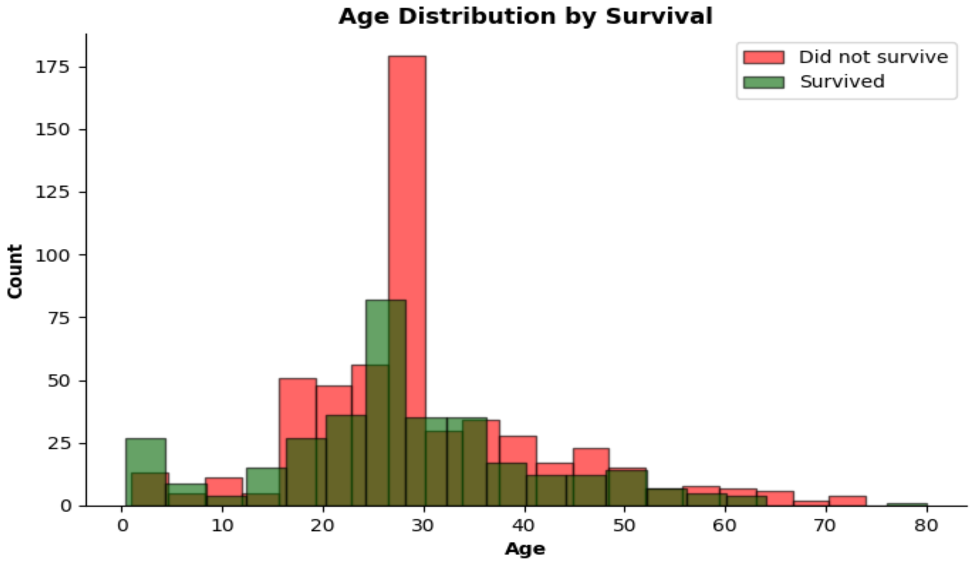
At a glance, it is obvious there was a priority system in the escape procedures, one which didn't only prioritize women and children as usual but also people of higher wealth classes. at least that's what the numbers show. As for the port of embarkation, it is clear that survival rates were more favourable for the people who embarked in Cherbourg. while there is yet to be further research done in the port-towns to know what makes Cherbourg stand out, one may assume it is a port which resides more affluent people.

However, it is important to note that the total survival rate was devastatingly low with numbers showing only a 38.38% survival rate.



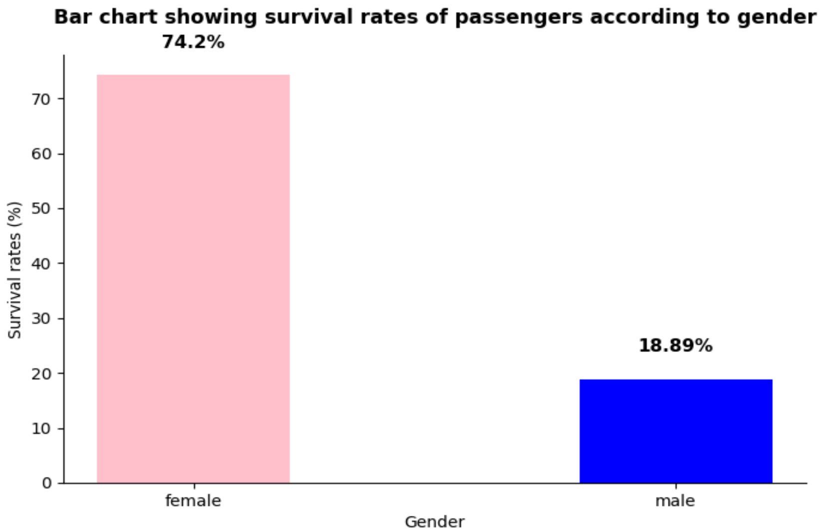
Finding 1: Children were given priority

While children had the highest survival rate at 57.97%, adults represented the largest absolute number of survivors due to their significantly higher representation in the passenger population.
The concentration of deceased passengers in the 20-35 age range reflects both the demographic makeup of the passenger population and the intersection of gender and class biases in the evacuation procedure.



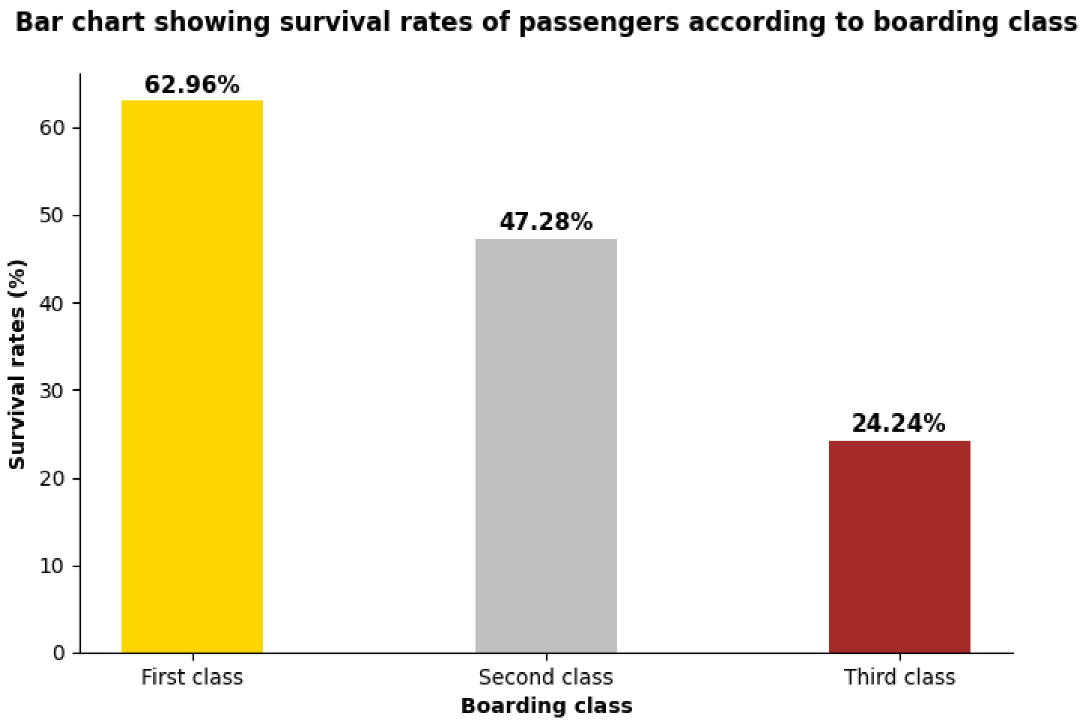
Finding 2: Gender was an influencing factor to survival

Cross-tabulation of gender and class reveals that gender was the dominant survival factor — 3rd class females (50%) had higher survival rates than 1st class males (36.89%), suggesting the 'women first' evacuation protocol overrode class privilege for male passengers.



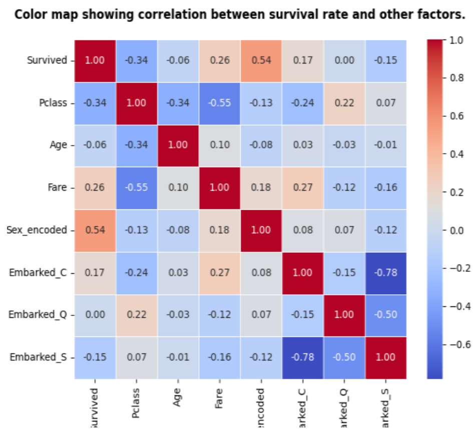
Finding 3: Passenger Wealth and Class also influenced their Survival.

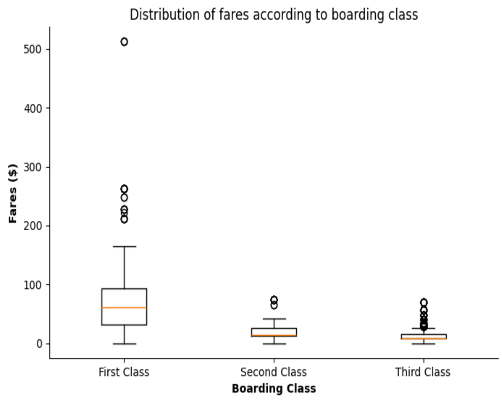
Statistical analysis also revealed a second priority system which favoured passengers of a higher boarding class; of all the people on the titanic, passengers who boarded with a higher-class boarding ticket were shown a more favourable survival rate than passengers with a lower-class boarding ticket.



Passengers who embarked at Cherbourg had a higher survival rate (55.36%) — likely because Cherbourg attracted wealthier passengers who disproportionately boarded in 1st class.

Also, fare distribution confirms significant wealth disparity between classes — 1st class median fares were approximately 6-8 times higher than 3rd class, supporting the observed correlation between class and survival rate."





Future evacuation procedures should ensure that priority is only focused on physical and mental vulnerabilities and should disregard socioeconomic standings such as wealth status.

Statistical confirmation.

Several statistical techniques were carried out on the dataset; one included a T test confirms the survival difference between genders is statistically significant ($p \approx 0$) — the probability of this difference occurring by chance is essentially zero. T-test also confirmed the survival difference between 1st and 3rd class passengers is statistically significant ($p \approx 0$) — wealth and social class had a definitive and measurable impact on survival probability

Bayesian analysis reveals that despite representing approximately 35% of passengers, females accounted for 68.13% of survivors — confirming that gender was the single most dominant factor in survival probability.

Conclusion and recommendations

The analysis reveals a clear hierarchy of survival on the Titanic — gender was the single most dominant factor, followed by passenger class and age. The data strongly suggests that evacuation procedures systematically favoured women, children and wealthy passengers. For historical researchers and policy makers studying disaster response, these findings highlight the critical importance of standardised, equitable evacuation protocols that prioritise need over social status.

Note on encoding: categorical variables were converted to numerical values for correlation analysis. Sex was encoded as male=0, female=1. Embarked was encoded as Southampton=0, Cherbourg=1, Queenstown=2